

This year's memorable relay race

The 52nd CERN relay race took place on the Meyrin site on 30 May. In total, 167 teams assembled on the starting line – a new record!



(Image: CERN)

More than 1000 runners, competing in 167 teams, took part in the CERN relay race on 30 May – a record level of participation for a highly memorable event! The race times ranged from 12 minutes and 15 seconds (the fourth best performance ever for the current route) to more than 23 minutes.

The Nordic walking race attracted 26 participants. And the children from the [Jardin des Particules](#) joined in the fun!

The [CERN Alumni](#) Network also took part in the event for the fourth year in a row by organising a [virtual relay race](#), in which 14 teams of alumni from a total of 26 countries competed.

The [photos](#) and the [results](#) of the relay race (worth consulting for the team names alone!) are now online on the Running club's website.

The CERN relay race is organised every year by the CERN Running club and the Staff Association. A big thank you to all the participants and volunteers, who make this race such a success.

Anais Schaeffer

A Word from Mike Lamont

Coming together to prepare for ambitious future projects

The first Accelerator and Technology sector workshop on 25 June will help to share expertise, challenges, methodologies and solutions

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A Word from Mike Lamont

Coming together to prepare for ambitious future projects

The first Accelerator and Technology sector workshop on 25 June will help to share expertise, challenges, methodologies and solutions

From concept to deployment, CERN's engineering practices emphasise precision, longevity, quality and cost management, enabling groundbreaking technology while maintaining high standards throughout the product lifecycle. The result is a truly remarkable level of performance across a diverse landscape of specialities.

Engineering processes at CERN are essential for creating the complex equipment that forms the backbone of the Laboratory. By integrating advanced fabrication tools, precision machining and rigorous quality control, we ensure that our components perform reliably for decades, even under extreme conditions. Our collaboration with external partners and our strong focus on innovation drive the development of sophisticated technologies.

All of this expertise is crucial as CERN prepares for an ambitious slate of projects. The high-luminosity upgrade of the LHC (HL-LHC) is set for deployment during Long Shutdown 3, which will start at the end of 2025. The high-intensity programme at the ECN3 facility is planned for the 2030s, alongside the ongoing two-phase North Area consolidation project. Additionally, the Future Circular Collider Feasibility Study will move to its next phase in 2025. These initiatives highlight the dynamic and forward-looking agenda of the Accelerator and Technology sector (ATS) and its four departments: Beams (BE), Engineering (EN), Accelerator Systems (SY) and Technology (TE).

The [first Accelerator and Technology sector](#) workshop is scheduled for 25 June at the CERN

Science Gateway. Targeting ATS sector members but open to the broader CERN community, the workshop will showcase the extensive range of engineering and technical activities in place across the sector. It will provide a platform for sharing experiences, expertise, challenges, methodologies and solutions, covering engineering design tools, processes and project management methodologies.

The workshop will be divided into two sessions, the first focusing on engineering design tools and processes, covering topics from digital twins to lifecycle management, and the second concentrating on project management methods and tools for future projects. This event offers a chance to explore various engineering and technical activities, network with colleagues across different teams and departments and build connections, whether over coffee or a beer at the end of the day. [Registration is now open.](#)

The recent [2023 engagement survey](#) showed the CERN community's strong commitment to working here, in this unique and challenging environment. However, it also highlighted the need for better collaboration across teams, improved recognition of achievements, more effective workload management and a greater focus on the future. This first ATS workshop is one of several measures being pursued to enhance collaboration, share insights and explore potential synergies across teams.

Mike Lamont

Director for Accelerators and Technology

Accelerator Report: Setting the stage for a productive summer

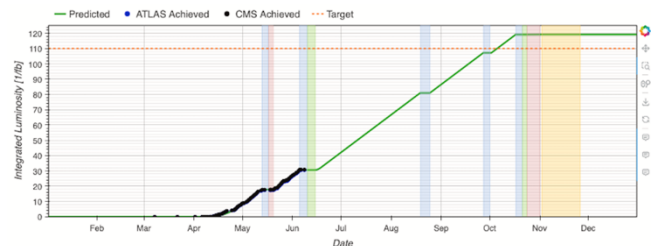
We are optimistic that we will have a successful luminosity production period this summer

On 5 June, the first period of luminosity production at the LHC came to an end and we began our second machine development (MD) session. The MD studies continued until 10 June,

when the last MD beams were dumped ahead of the scheduled technical stop. Luminosity production is expected to resume on 17 June for a period of nine weeks of uninterrupted production,

until the next MD session, which is scheduled to start on 19 August.

When production beams were dumped on 5 June, the counter for the integrated luminosity showed 30.4 fb^{-1} for ATLAS and CMS, 2.5 fb^{-1} for LHCb and 17.3 fb^{-1} for ALICE, exactly as expected.



The scheduled (green line) and achieved (black dots) integrated luminosity for ATLAS and CMS for the 2024 proton–proton run. The blue shaded areas indicate the MD sessions, the green shaded areas indicate the technical stops, the red shaded areas represent special physics runs and, finally, the yellow shaded area represents the period with heavy ion collisions. (Image: CERN)

During the MD session, many important and very interesting topics were studied, which will benefit both this year’s run and future operations. Notably, the machine optics for the High-Luminosity LHC (HL-LHC) were studied to preempt any issues and ensure that we have enough time to address them.

In the [Accelerator Report of 3 May](#), I discussed the collimator hierarchy problem that emerged on 17 April, when the machine was filled with 1791

bunches per beam. This issue limited the squeeze process, slightly reducing luminosity production. Following numerous studies and a final test last week, the LHC team identified several corrections to be applied to the machine’s settings, which will require it to be revalidated. This will have no impact on physics time, since a short intensity ramp-up is always performed after a technical stop in order to revalidate the accelerator. Once validation is complete, the squeeze process can go back to the values initially planned for this year’s run.

The 2024 run has been extended by four weeks of proton physics, as mentioned in my [Accelerator Report of 16 April](#). The 2024 target for the integrated luminosity has thus been updated from 90 fb^{-1} to 110 fb^{-1} .

With the extra knowledge and experience gained from the MD session, the preventive maintenance performed on the machine and the resolution of the collimator hierarchy issue, we are optimistic that we will have an uninterrupted and successful luminosity production period this summer. We aim to reach an integrated luminosity of 80 fb^{-1} for ATLAS and CMS before the next MD session starts, on 19 August, on the way to our new 2024 luminosity target of 110 fb^{-1} .

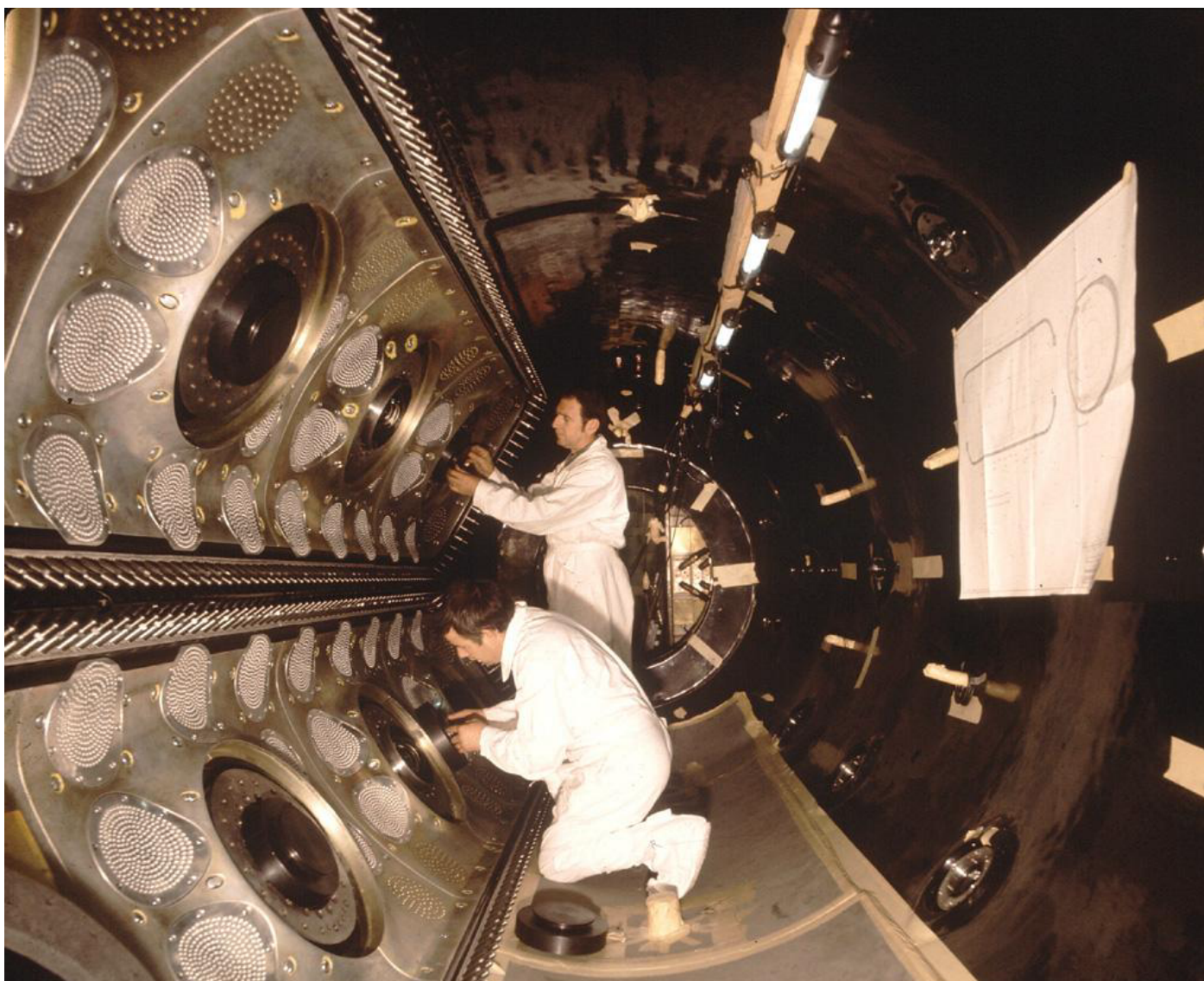
Rende Steerenberg

CERN70: A gargantuan discovery

Part 11 of the [CERN70](#) feature series. Violette Brisson played an active part in the discovery of neutral currents; she was head of the Gargamelle group at the Laboratory of the Ecole Polytechnique in Paris

It was the first great discovery to be made at CERN. On 19 July 1973, the [Gargamelle](#) group announced that they had found evidence of weak neutral currents. It was a groundbreaking discovery that paved the way towards a grand unification theory and the associated new physics. According to this theory, at the high energy levels that existed just after the birth of the Universe, all forces were unified and only became distinct as the Universe cooled.

As early as the 19th century, James Maxwell had demonstrated that the electrical and magnetic forces were just two manifestations of the same interaction. From the ferment of new theories developed in the 1960s emerged the concept that the weak force could be unified with the electromagnetic force as a single interaction, the so-called electroweak interaction. The theory predicted the existence of a specific manifestation of the weak force, referred to as weak neutral currents. Gargamelle was the first experiment to find proof of these currents, thereby confirming the electroweak theory.



Inside the Gargamelle bubble chamber in 1970 (Image: [CERN](#))

The Gargamelle experiment was outlined in 1963 by physicist André Lagarrigue and named after the mother of Gargantua, the giant in François Rabelais' story. It was a huge bubble chamber, weighing 1000 tonnes and containing 18 tonnes of liquid freon. The chamber's large volume, for its time, enabled the interaction of neutrinos – particles with no electric charge that interact only weakly. Built by laboratories at the French Atomic Energy Commission (CEA), the Ecole Polytechnique and the Faculty of Science in Orsay, Gargamelle eventually gathered a team of about 50 physicists. The experiment began operation in January 1971, and delivered a first significant result when it confirmed that the proton consists of several particles, the "[quarks](#)". But its greatest achievement was providing evidence of neutral currents in 1973.

Recollections

Head of the Gargamelle group at the Laboratory of the Ecole Polytechnique in Paris, of which she was deputy director from 1973 to 1985, Violette Brisson played an active part in the discovery of neutral currents. She joined the Linear Accelerator Laboratory in Orsay in 1988, and the Franco-Italian VIRGO experiment in the 1990s.

"I couldn't begin this account without mentioning my colleagues from the Ecole Polytechnique who

are no longer with us: André Lagarrigue, without whom there would be no story to tell, André Rousset, who was in charge of the CERN group at that time and Paul Musset, who also played a very important role in the discovery.

Neutral currents, whose existence was extremely hypothetical in those days, were only eighth on the planned list of experiments for Gargamelle. It was not until 1972 that the theorists drew the Collaboration's attention to the work of Gerard 't

Hooft, who had predicted their existence in the framework of the new theories unifying the electromagnetic and weak forces.

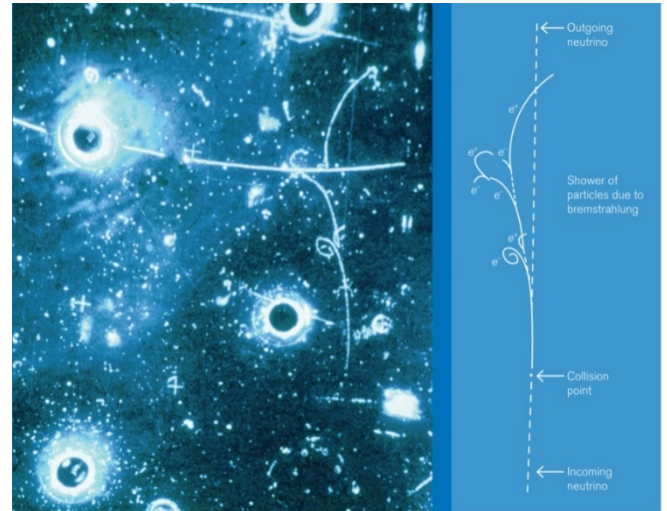


Violette Brisson, a highly respected member of the French particle physics community, played a leading role in the discovery of neutral currents. (Image: [École Polytechnique](#))

The associated research was initiated by Antonino Pullia in Milan, who very soon presented us with about 30 possible events, or "candidates", which generated a lot of excitement. Several physicists, including myself at the Ecole Polytechnique, together with Pierre Petiau and Louis Kluberg, threw themselves into research on the subject. We very quickly found candidate events for neutral currents but I was still very doubtful until a meeting at CERN in May 1972, when all the laboratories involved in the Collaboration presented photos of their candidates. I came away firmly convinced, as did many of my colleagues. That was when we really began to get excited because, after finding so many semi-precious stones in our experiments over the past ten years, we were now starting to glimpse a genuine diamond!

This marked the start of an intense phase of work, which involved sorting through some 100 000 photos and analysing events, as well as studying their background (since neutron interactions can be confused with neutral currents). In December 1972, there was great rejoicing when virtually incontestable proof of another type of neutral current, the "leptonic" neutral current, was found in Aachen. The theorists were ecstatic. We re-examined the pictures (four times!) to make sure that we hadn't missed anything. Later on, our analysis of nine times the number of shots

produced only two further candidates, which tied in with the theoretical calculations.



The first example of a single-electron neutral current. An invisible incoming antineutrino knocks an electron forward creating a track that seems to appear from nowhere. (Image: [CERN](#))

We were euphoric as this confirmed the existing evidence for so-called "hadronic" neutral currents. However, extensive work was needed at the beginning of 1973 to calculate the background for these hadronic currents. Once the simulation programmes of Dieter Haidt and Jean-Pierre Vialle and the more simple calculations of André Rousset, Don Perkins and Antonino Pullia, to name but a few, had been completed, there was no longer any shadow of a doubt. With the unanimous agreement of the whole Collaboration, Paul Musset presented the results at CERN in July 1973 and they were published in early September. Despite the enthusiasm of the entire community, some of our colleagues remained sceptical. We ourselves were confident, however, especially since the American team at Fermilab had also found candidates. That confidence lasted until one day in October 1973, when André Lagarrigue received a telephone call informing him that Fermilab's candidates had disappeared. I can still remember the anguish in his voice when he called to tell me that our colleagues "over there" could no longer see anything and wondered whether we could have forgotten something or made a mistake. Doubts began to creep in...

In order to convince our detractors, we took data that checked the background results obtained by Dieter Haidt. We even allowed ourselves to make jokes about our American colleagues' "alternating

currents". All doubt was finally removed with the discovery of two further leptonic neutral currents at the beginning of 1974. The Americans' neutral currents gradually re-emerged and, in summer 1974, the existence of the phenomenon was officially recognised.



Gargamelle's bubble chamber is now on display at [Square Van Hove](#), in the grounds of CERN's Science Gateway. (Image: [CERN](#))

The discovery gave rise to many neutrino experiments at CERN and in the United States, which measured all the parameters of the

electroweak interaction with increasing precision. The discovery of the [W](#) and [Z](#) bosons at CERN in 1983 then confirmed the electroweak theory once and for all.

Unfortunately, Gargamelle was not destined to ring the Nobel chimes for André Lagarrigue, the "father" of the experiment, who passed away in 1975 before he could be considered for the Prize. Having been "put to rust" for several years, the bubble chamber was finally erected at CERN as a piece of sculpture. As for Gargamelle's "workers", we were sometimes badly treated or forgotten, but we also experienced the elation and joy of a very great discovery and we have always remained true friends. What more could anyone ask for?"

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. Violette Brisson [died in 2018](#) at the age of 83. In 2023, CERN [celebrated](#) the 50th anniversary of the discovery of neutral currents, and the CERN Courier published "[CERN's neutrino odyssey](#)" charting CERN's neutrino programme since Gargamelle.

Using carbon dioxide to reduce carbon dioxide emissions

Find out how CERN is minimising its carbon dioxide footprint with a novel approach using carbon dioxide cooling technology



Part of the new cooling system, installed on the roof of CMS (LHC Point 5, Cessy). (Image: CERN)

For [World Environment Day](#), celebrated by the United Nations on 5 June, CERN reaffirms its commitment to [environmentally responsible research](#). Among numerous actions, CERN has a dedicated strategy to reduce emissions, which targets gas recirculation, gas recovery and

exploring the use of alternative gases. Currently, the majority of CERN's direct greenhouse gas emissions come from its particle detectors, which use a range of gas mixtures for particle detection and detector cooling. These gases are mainly synthetic refrigerants, including fluorinated gases with a particularly high global warming potential. Since 2017, CERN has been developing a [novel approach](#) to detector cooling using carbon dioxide (CO₂). CO₂ has a global warming potential of 1, which is several thousand times lower than the synthetic refrigerants currently used in low-temperature refrigeration systems, making it an excellent alternative. The Engineering department's Cooling and Ventilation group and the Experimental Physics department's Detector Technology group, with the support of other

teams across CERN and partners in science and industry, are currently renovating the cooling systems of the [ATLAS](#) and [CMS](#) inner detectors. Surface work is already under way, while underground work will take place during the next long shutdown, LS3, scheduled to begin at the end of 2025. The objective is to achieve a drastic reduction of direct emissions of fluorinated gases, saving the equivalent of 40 000 tonnes of CO₂ each year.

How? Every technical parameter has been optimised to cool CO₂ to -53 °C, close to the temperature where CO₂ becomes solid (-56.6 °C), pushing the performance of the equipment and the standard cooling cycles. Not only will this technology contribute to CERN's objective of reducing its emissions, but it could also have applications in other low-temperature industries, notably the food and pharmaceutical industries,

furthering CERN's tradition of knowledge and technology transfer for the benefit of society. Find out more in the new video below.

CO₂ Cooling upgrade for the main detectors (Video: CERN)

This is just one of CERN's initiatives to minimise its impact on the environment in key domains, including energy, water, waste, sustainable land use, noise and emissions.

Find out more in the new video: <https://videos.cern.ch/record/2300177>.

This is just one of CERN's initiatives to minimise its impact on the environment in key domains, including energy, water, waste, sustainable land use, noise and emissions. Find out more on: <https://home.cern/about/what-we-do/environmentally-responsible-research>.

HSE unit

Future colliders and fusion reactors

CERN and EUROfusion are collaborating to develop innovative technologies for future colliders and nuclear fusion reactors



This high-temperature superconducting (HTS) cable, developed at CERN, is inspired by those developed for advanced nuclear fusion reactor concepts. It consists of a hollow copper profile with twisted grooves containing stacks of soldered HTS tapes (shown opened in a "flower" shape in the image). (Image: CERN)

CERN's accelerator experts and EUROfusion's nuclear fusion specialists are now working jointly to develop innovative technologies for future colliders and nuclear fusion reactors, drawing on their respective unique competencies, in particular in the area of high field magnets.

The common projects are facilitated by the collaboration agreement that was signed in November 2023 by CERN and members of [EUROfusion](#), the European consortium of fusion research laboratories carrying out a [technical design of a fusion demonstration power plant \(DEMO\)](#) to succeed [ITER](#).

Marking a milestone in scientific cooperation, this partnership paves the way for joint ventures in a broad spectrum of areas, encompassing research and development in physics, engineering and technology. It focuses on the engineering design and construction of significant scientific experiments and instruments.

CERN established a dedicated fusion technology coordination unit in 2023 with the involvement of accelerator and [knowledge transfer](#) experts, reflecting the Organization's commitment to fostering collaboration across multiple scientific disciplines. The agreement signed between EUROfusion and CERN is a prominent example of the unit's activities.

“There are clear synergies in the development of technologies for nuclear fusion and those for future colliders, particularly with regard to the use of high-temperature superconducting (HTS) conductors. In the ideation of DEMO, the demonstration fusion power plant that will succeed ITER, the choice of tokamak central solenoids using HTS materials is being explored by a number of EUROfusion members”, says Gianfranco Federici, Head of the EUROfusion Fusion Technology Department. “The collaboration agreement provides a platform for all of our members and CERN to collaborate and create a centre of excellence for fusion technology”.

Central to this alliance, the inaugural project initiated this year is envisioned as a crucial step towards a future testbed for tritium blanket technology. It is scheduled for completion by the end of 2024. “The teams at CERN and EUROfusion are engaging in fruitful exchanges concerning

magnet concepts and designs based on high-temperature superconductors. The extraordinary challenges posed by the magnets of a muon collider require the development of new concepts, some of them similar to those of fusion machines. This is particularly true for the target solenoid, a key component of the collider, which needs to reach a very high field, is subjected to large heat and radiation loads and is likely to be built with HTS conductors”, says Luca Bottura, who is leading the magnet design efforts of the International Muon Collider Collaboration and the magnet work package of the [EU MuCol design study](#).

Recently, the scope of the collaboration has been extended to include cooperation on advanced shielding materials, of interest both to fusion and accelerators. This topic will be the focus of a joint workshop, which is due to take place later this year.

Antoine Le Gall & Anaïs Schaeffer

CERN welcomes International Year of Quantum Science and Technology

On the centenary of quantum mechanics -- the bedrock of particle physics and enabler of numerous technologies – CERN is contributing to the development of a new generation of quantum technologies for fundamental research and beyond



View of CERN Data Centre. (Photo: Roger Claus/CERN)

100 years ago, a handful of visionary physicists upturned notions about nature that had guided scientists for centuries. Particles can be point- or wave-like, depending on how you look at them. Their behaviour is probabilistic and can momentarily appear to violate cherished laws

such as the conservation of energy. Particles can be entangled such that one feels the change of state of the other instantaneously no matter the distance between them, and, as befalls Schrödinger's famous cat, they can be in opposite states at the same time.

Today, thanks to pioneering theoretical and experimental efforts to understand this complex realm, physicists can confidently navigate through such apparently irrational concepts. Quantum theory has not only become foundational to physics, chemistry, engineering and biology, but underpins the transistors, lasers and LEDs that drive modern electronics and telecommunications -- not to mention solar cells, medical scanners and global positioning systems. But this is only the beginning.

On 7 June the United Nations declared 2025 the [International Year of Quantum Science and](#)

[Technology](#) to celebrate the contributions of quantum science to technological progress, raise awareness of its importance to sustainable development, and ensure that all nations have access to quantum education and opportunities. As the world's largest particle physics lab, CERN has been interrogating the quantum theories that govern the microscopic world for the past 70 years. Most recently, it has entered the rapidly growing domain of quantum technologies, which aims to harness the strangest aspects of quantum mechanics to build a new generation of quantum devices for fundamental research and beyond.

"In recent years, we have learned not just to use the properties of the quantum world but, also, to control them," explains Sofia Vallecorsa, coordinator of the CERN Quantum Technology Initiative (QTI). "Today, the revolution is all about controlling individual quantum systems, such as single atoms or ions, enabling even more powerful applications."

At CERN, quantum technologies are studied and developed through two initiatives: the [QTI](#), whose aim is to enable technologies – such as quantum computing, quantum state sensors, time synchronisation protocols, and many more – for high-energy physics activities; and the recently established Open Quantum Institute ([OQI](#)), whose aim is to identify, support and accelerate the

development of future societal applications benefiting from quantum computing algorithms. One of the most promising fields is quantum computing. Unlike conventional computers that use "bits" that can be in one of just two states, quantum computers using qubits which can exist in superpositions of states. This enables a vast number of computations to be processed simultaneously, offering important applications in fields such as cryptography, logistics and process optimisation, and drug discovery. Quantum communication, which exploits the principles of quantum mechanics to make it impossible to intercept information without detection, is another significant area of development. A third pillar of CERN's quantum-technologies programme is sensing to allow ultra-precise measurements of physical quantities, with potential applications in fields including medicine, navigation and climate science.

"What started 100 years ago as a purely theoretical physics investigation is now beginning to unleash its full potential," says OQI coordinator Tim Smith of CERN. "The International Year of Quantum Science and Technology will be a wonderful opportunity to celebrate the past, the present and the future of our understanding of the quantum world."

Antonella Del Rosso

Shaking the box for new physics

The CMS collaboration has searched for new physics in a rare decay of a known particle, using an approach that can be likened to shaking a box containing a birthday present to find a clue about what's inside

When you receive a present on your birthday, you might be the kind of person who tears off the wrapping paper immediately to see what's inside the box. Or maybe you like to examine the box, guessing the contents from its shape, size, weight or the sound it makes when you shake it.

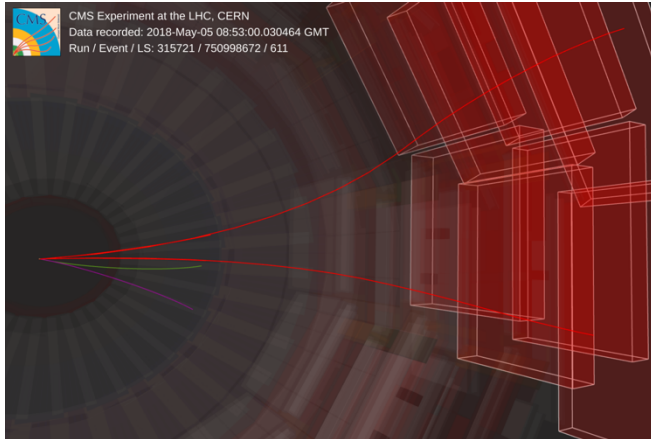
When physicists at the [Large Hadron Collider](#) (LHC) analyse their datasets in search of new physics phenomena such as new particles, they usually take one of two different approaches. They either perform a direct search for a specific new kind of particle, equivalent to tearing off the wrapping paper immediately, or use an indirect strategy

based on quantum mechanics and its subtle wonders, similar to shaking the box and guessing what's inside.

At the annual [LHCP conference](#) that took place in Boston last week, the [CMS](#) collaboration reported how it used the second approach to look for new physics in a rare decay of a particle called B^0 meson.

The physics process that drives the decay of a particle into lighter ones can be influenced by new, unknown particles, which might be too heavy to be produced at the LHC. This influence could change the decay process in ways that can be

measured and compared to predictions of the [Standard Model](#) of particle physics. In the same way as shaking the box containing your birthday present could give you a clue about what's inside, any deviation from the Standard Model predictions could give physicists a hint of new physics.



*CMS candidate collision event for a B^0 meson decaying into a K^{*0} meson and two muons (red lines). The K^{*0} meson decays into a K^+ meson (magenta line) and a π^- meson (green line). (Image: CERN)*

The decay of the B^0 meson, which is made up of a bottom quark and a down quark, into a K^{*0} meson (containing a strange quark and a down quark) and two muons is particularly suited to this approach. This is because it occurs via a rare [penguin](#) transition that is highly sensitive to possible contributions from new heavy particles.

In its new study, the CMS team used all the data collected by its detector between 2016 and 2018, during the second run of the LHC, to “shake” this B^0 decay “box”. This box offers many ways to look for new physics. One is to weigh the box, i.e.

measure the rate at which the decay occurs. Another is to take two twin boxes – for example, one corresponding to the decay into two muons and the other to the decay into two electrons – and check if they weigh the same.

In their new study, the CMS researchers looked at the shape of the box, i.e. they examined how the particles produced in the decay share the energy of the parent B^0 meson and measured at what angles they fly away from each other. They then determined a set of parameters using these energies and angles, and compared the results with two sets of predictions from the Standard Model.

For most parameters, the results are in line with these two sets of Standard Model predictions. However, for two parameters, known as P_5' and P_2 , and for specific energies of the two muons, the results are in tension with the two available predictions. Overall, the results are in agreement with the previous results from the ATLAS, LHCb and Belle experiments, while improving upon their level of precision.

Unfortunately, there is a charming, “naughty” kind of penguin that’s crashing the birthday party: a charm quark that participates in the rare penguin transition. This complicates the Standard Model predictions and makes it difficult to draw a conclusion. To advance, researchers need better predictions, more data and improved analysis techniques.

Find out more on the CMS website: <https://cern.ch/oya24>.

CMS collaboration

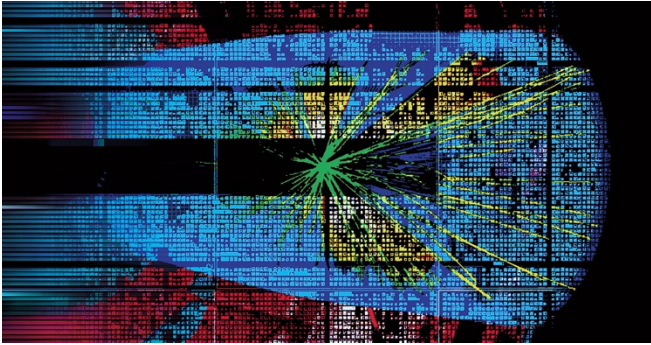
How can AI help physicists search for new particles?

The ATLAS and CMS collaborations are using state-of-the-art machine learning techniques to search for exotic-looking collisions that could indicate new physics

One of the main goals of the LHC experiments is to look for signs of new particles, which could explain many of the unsolved mysteries in physics. Often, searches for new physics are designed to look for one specific type of new particle at a time, using theoretical predictions as a guide. But what

about searching for unpredicted – and unexpected – new particles? Trawling through the billions of collisions that occur in the LHC experiments without knowing exactly what to look for would be a mammoth task for physicists. So, instead of sifting through the data and looking for anomalies,

the ATLAS and CMS collaborations are letting artificial intelligence (AI) do the job.



(Image: S Sioni/CMS-PHO-EVENTS-2021-004-2/M Rayner)

At the *Rencontres de Moriond* conference on 26 March, physicists from the CMS collaboration presented [the latest results](#) obtained by using various machine learning techniques to search for pairs of “jets”. These jets are collimated sprays of particles originating from strongly interacting quarks and gluons. They are particularly difficult to analyse, but they could be hiding new physics.

Researchers at ATLAS and CMS use several strategies to train AI algorithms in their searches for jets. By studying the shape of their complex energy signatures, scientists can determine what particle created the jet. Using real collision data, physicists at both experiments are training their AI to recognise the characteristics of jets originating from known particles. The AI is then able to differentiate between these jets and atypical jet signatures, which potentially indicate new interactions. These would show up as an accumulation of atypical jets in the data set.

Another method involves instructing the AI algorithm to consider the entire collision event and look for anomalous features in the different particles detected. These anomalous features may indicate the presence of new particles. This technique was demonstrated in a paper released by ATLAS in [July](#) 2023, which featured one of the first uses of unsupervised machine learning in an LHC result. At CMS, a different approach involves physicists creating simulated examples of potential new signals and then tasking the AI with identifying collisions in the real data that are different to regular jets but resemble the simulation.

In the latest results presented by CMS, each AI training method exhibited varying sensitivities to different types of new particles, and no single algorithm proved to be the best. The CMS team was able to limit the rate of production of several different types of particles that produce anomalous jets. They were also able to show that the AI-led algorithms significantly enhanced the sensitivity to a wide range of particle signatures in comparison to traditional techniques.

These results show how machine learning is revolutionising the search for new physics. “We already have ideas about how to further improve the algorithms and apply them to different parts of the data to search for several kinds of particles,” says Oz Amram, from the CMS analysis team.

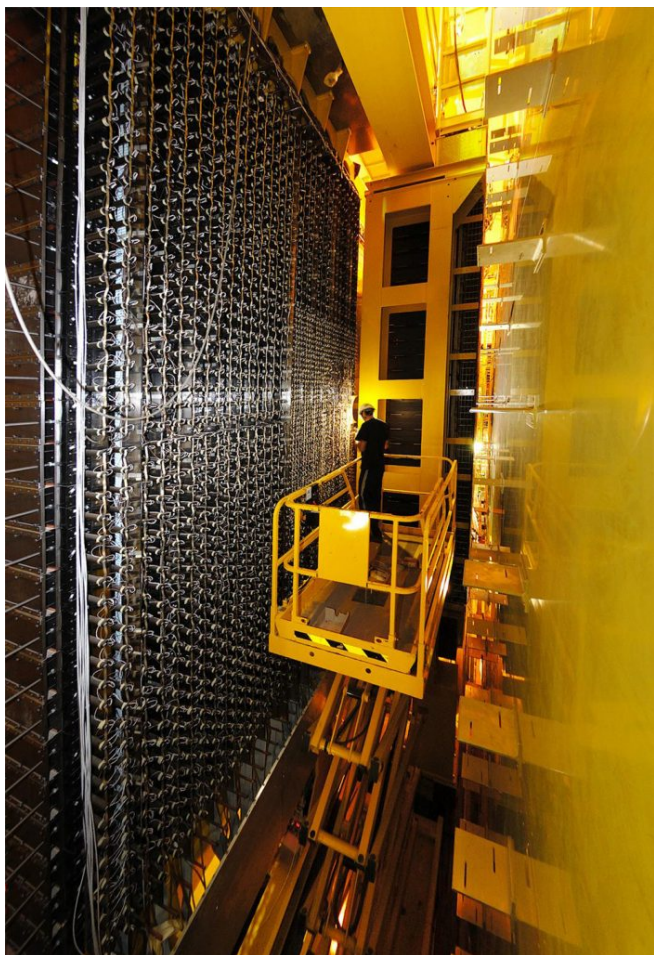
CMS collaboration

Upgrading the LHCb sub-detectors for the HL-LHC

The LHCb experiment is revamping its electromagnetic calorimeter with new high-performance modules and equipping its ring-imaging Cherenkov detectors with very fast electronics

On 5 July 2022, protons began colliding again in the LHCb detector after a three-and-a-half-year break known as Long Shutdown 2 (LS2), marking the start of the third run of the Large Hadron Collider (LHC). During this period, the original LHCb detector at the LHC was largely dismantled and an almost completely new detector constructed. The 2020 update of the European Strategy for Particle Physics approved by the CERN Council strongly supported exploiting the full

potential of the LHC for studying flavour physics. A further upgrade of the LHCb detector, known as Upgrade II, is planned to allow LHCb to operate at a much higher instantaneous luminosity and cope with the demanding data-taking conditions of the High-Luminosity LHC (HL-LHC). The latest technological developments will be taken into account to design the new detectors.



Maintenance of the electromagnetic calorimeter (Image: LHCb collaboration)

The new revolutionary electromagnetic calorimeter being developed for LHCb Upgrade II will be able to precisely measure the arrival time of electromagnetic particles. Its test measurements demonstrated detection of high-energy electrons within 20 picoseconds of precision. This is the first time that such excellent performance has been achieved with an electromagnetic calorimeter in particle physics. The new calorimeter will have finer granularity, enabling it to cope with a much higher particle density at higher instantaneous luminosity. Luminosity is an important indicator of the performance of an accelerator: it is proportional to the number of particle collisions that occur in a given amount of time.

The technology for the new LHCb calorimeter modules is based on the “Spaghetti Calorimeter”

(SpaCal) concept, in which the scintillators resemble strands of spaghetti. These scintillating fibres are housed in the 5180 longitudinal holes in the SpaCal module. The calorimeter is constructed with modules with two types of absorbers: tungsten-based SpaCal-W modules, which will be built using tungsten 3D-printing technology, and lead-based SpaCal-Pb modules.

The number of particles crossing the detector is extremely high in the central region surrounding the beam pipe, inside which the proton beams of the LHC circulate. The current LHCb calorimeter is composed of Shashlik modules. The performance of these modules decreases over time due to radiation damage caused by the large flux of particles when the LHC is running. During the next Long Shutdown, these degraded Shashlik modules in the very central region around the LHC beampipe will be replaced with 32 SpaCal-W modules, and an additional 144 SpaCal-Pb modules will be placed around them.

In the upgrade of the ring-imaging Cherenkov (RICH) system, the whole electronics chain will be replaced. The detector will be equipped with a high-rate data acquisition system and a novel readout application-specific integrated circuit, called FastRICH, which is capable of providing precise timestamps of Cherenkov photons. These photons are produced when electrically charged particles, such as protons or electrons, travel faster than light in a medium.

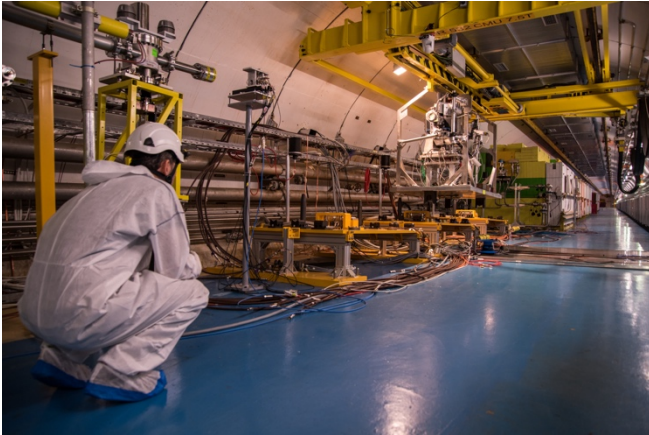
The new LHCb RICH will be the first system featuring fast timing capabilities for single photons at the hundred-picosecond level, demonstrating once again how the challenging conditions of a flavour physics experiment at the LHC can lead to technological breakthroughs.

Prototypes of both subdetectors, SpaCal and RICH, have been successfully tested at CERN’s accelerator complex with beams from the Super Proton Synchrotron in preparation for the HL-LHC era.

LHCb collaboration

Bringing black hole jets down to Earth

The Fireball collaboration has used CERN's HiRadMat facility to produce an analogue of the jets of matter and antimatter that stream out of some black holes and neutron stars



View of the HiRadMat facility (Image: CERN)

Dive into the heart of an active galaxy and you'll find a supermassive black hole gobbling up material from its surroundings. In about one out of ten such galaxies, the black hole will also shoot out jets of matter at close to the speed of light. Such relativistic black hole jets are thought to contain, among other components, a plasma of pairs of electrons and their [antimatter](#) equivalents, positrons.

This relativistic electron–positron plasma is believed to shape the dynamics and energy budget of the black hole and its environment. But how exactly this happens remains little understood, because it's difficult both to measure the plasma with astronomical observations and to simulate it with computer programmes.

In a [paper](#) just published in *Nature Communications*, Charles Arrowsmith and colleagues from the Fireball collaboration report how they have used the [HiRadMat](#) facility at CERN to produce a relativistic beam of electron–positron plasma that allows this medium to be studied in detail in laboratory experiments.

Relativistic beams of electron–positron pairs can be created in several ways at different types of laboratories, including high-power laser facilities. However, none of the existing ways can produce the number of electron–positron pairs that is required to sustain a plasma – a state of matter in which the constituent particles are very loosely connected. Without sustaining the plasma, researchers cannot investigate how these analogues of black hole jets change as they move

through a laboratory equivalent of the interstellar medium. This investigation is key to explaining observations from ground- and space-based telescopes.

Arrowsmith and colleagues found a way to meet these requirements at CERN's HiRadMat facility. Their approach involved extracting within a mere nanosecond a whopping three hundred billion protons from the Laboratory's [Super Proton Synchrotron](#) and firing them onto a target of graphite and tantalum, in which a cascade of particle interactions generates huge numbers of electron–positron pairs.

By measuring the resulting relativistic electron–positron beam with a set of instruments, and comparing the result with sophisticated computer simulations, Arrowsmith and co-workers showed that the number of electron–positron pairs in the beam – more than ten trillion – is ten to hundred times greater than previously achieved, exceeding for the first time the number needed to sustain the plasma state.

“Electron–positron plasmas are thought to play a fundamental part in astrophysical jets, but computer simulations of these plasmas and jets have never been tested in the laboratory,” says Arrowsmith. “Laboratory experiments are necessary to validate the simulations, because what seems like reasonable simplifications of the calculations involved in the simulations can sometimes lead to drastically different conclusions.”

The result is the first from a series of experiments that the Fireball collaboration is carrying out at HiRadMat.

“The basic idea of these experiments is to reproduce in the laboratory the microphysics of astrophysical phenomena such as jets from black holes and neutron stars,” says co-author of the paper and lead researcher Gianluca Gregori. “What we know about these phenomena comes almost exclusively from astronomical observations and computer simulations, but telescopes cannot really probe the microphysics and simulations involve approximations. Laboratory experiments

such as these are a bridge between these two approaches.”

Next in Arrowsmith and colleagues’ plasma pursuits at HiRadMat is to have these powerful jets propagate through a metre-long plasma and observe how the interaction between them generates magnetic fields that speed up the particles in the jets – one the greatest puzzles in high-energy astrophysics.

“The Fireball experiments are one of the latest additions to HiRadMat’s portfolio,” says operation

manager of the facility Alice Goillot. “We’re looking forward to continue reproducing these rare phenomena using the unique properties of CERN’s accelerator complex.”

This project has received funding from the European Union’s Horizon Europe Research and Innovation programme under Grant Agreement No 101057511 (EURO-LABS).

Ana Lopes

LHC tunnel named as one of the 50 most iconic tunnels in the world

The LHC tunnel is one of the “most remarkable feats of civil engineering” according to the International Tunnelling Association



The LEP/LHC tunnel in 1985. Three tunnel-boring machines started excavating the tunnel in February 1985 and the ring was completed three years later. (Image: CERN)

Construction work for the tunnel of the world’s most iconic particle accelerator began in 1983. The tunnel was initially home to the Large Electron–Positron Collider, which ran from 1989 to 2000, and now houses the Large Hadron Collider. During construction of the tunnel, the team had to overcome multiple engineering challenges,

including boring a 27-km circular tunnel with a constant radius and dealing with numerous geological challenges. From 1998 to 2005, it was majorly upgraded to prepare it for the LHC. These civil engineering works included constructing new transfer tunnels from the SPS to the LHC and giant underground caverns to house the ATLAS and CMS detectors.

Now, this feat has been recognised by the International Tunnelling and Underground Space Association (ITA) as one of the 50 most iconic tunnels in the world, on the occasion of the Association’s 50th anniversary. The ITA has published a book to commemorate this list, which includes impressive projects such as the Gotthard Base Tunnel and the Channel Tunnel.

The list was revealed at the annual World Tunnel Congress in Shenzhen in April. Here, the civil engineering plans for the Future Circular Collider were presented to the global tunnelling community. Owing to its scale and technical complexity, the project has attracted a great deal of interest from designers and contractors, and the executive committee of the ITA even paid a special visit to CERN in December 2022. Read more about the FCC tunnel in the [CERN Courier](#).

John Osborne

Computer Security: Blind trust means money lost

We acknowledge that finding decent accommodation near CERN in the Geneva area or the Pays de Gex is difficult. Particularly difficult if you're trying to organise such accommodation from abroad. Demand wildly exceeds supply. And where customers are in (desperate) need of supply, fraudsters are never far away.

And indeed, this spring has seen two fraudulent transactions on the CERN Marketplace – your private flea market – a place to sell books, music or electronics, buy a new (used) car or find accommodation. In both cases, the fraudsters were advertising vacant apartments that they didn't own. Using fake identities, hardly traceable email addresses and even, in one case, a stolen passport, they engaged with potentially interested tenants and provided details, photos and a location. A seemingly plausible story. But a fake one. And they pushed their victims into signing a lease and talked them into paying a deposit – which can easily be a few thousand euros – money that was subsequently lost.

As with any other online market platform (Amazon, eBay, etc.) or real-life flea market, a sales/purchase contract is also an expression of trust. The seller trusts that the money you pay is genuine, that the transfer you make is non-revocable, that the cheque is covered. The buyer trusts that the seller will really hand over or ship the merchandise and that the merchandise is as described and without any unmentioned flaws or faults. This is why, usually, face-to-face transactions using cash are preferable. As a buyer, you can see (and test) what you buy. As a seller, you can take comfort in the fact that counterfeit money is hard to come by (while uncovered cheques are easily possible). And this is why, in the online world, payments should usually be made through a trusted partner – an escrow service like eBay offers them. Or transactions should be

backed up by insurance (like with Amazon or PayPal).

The CERN Marketplace is an unmoderated forum for private transactions – sales, purchases, etc. Other than providing the platform, CERN does not engage further nor take any responsibility for its contents. Participation just requires an email address as a handle. Those participants are neither vouched for nor vetted. In addition, while posts must comply with CERN's Computing Rules and the dedicated rules of the CERN Marketplace, they are generally not moderated, verified or approved. Of course, any violation of those rules will be followed up by the CERN Computer Security team and might lead to posts being deleted, participants being blocked, and sanctions (including calling the local police) – as happened in the aforementioned cases.

But don't take a chance. BE VIGILANT. Like in real life. Unless you know the seller/buyer, don't fully trust them. If you're looking for an apartment to rent, it's advisable to either seek input from your peers ("Does anyone know that landlord? Are they trustworthy?") or do an in-person visit, ask for a live video tour of the apartment or ask a local friend to do a visit for you. In any case, avoid transferring money to unknown parties. Perhaps set up an escrow account where money is transferred only under certain conditions. Remember: "Blind trust means money lost".

Computer Security team

Do you want to learn more about computer security incidents and issues at CERN? Follow our Monthly Report. For further information, questions or help, check our website or contact us at Computer.Security@cern.ch.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

27 May–27 June | At CERN | Meyrin and Prévessin | CERN Table Tennis Club | EN & FR

[**CERN 70th Anniversary Table Tennis Tournament**](#)

13 Jun 16:30 | Knowledge Sharing | Online | CERN Alumni Network | EN

[**Celebrate 7 years of the CERN Alumni Network**](#)

13 Jun 17:30 | Knowledge Sharing | Council Chamber | Staff Association | FR

[**« La femme naît libre et demeure égale à l'homme en droits », par Valentina Altopiedi**](#)

14 Jun 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[**L&D Micro-talk 11 – Cultivating Emotional Intelligence \(EI\) in the Era of Artificial Intelligence \(AI\)**](#)

14 Jun onwards | At CERN | R1 & R2 | Novae | FR
Euro 2024 matches on big screens, R2 at 15:00, R1 at 18:00 & 21:00

17 Jun 11:00 | Knowledge Sharing | 40/S2-C01 & Zoom | KT Seminars | EN

[**AI and computing challenges: harnessing Large Language Models at Thomson Reuters**](#)

25 Jun | Engineering | CERN Science Gateway | ATS sector | EN

[**Accelerator and Technology sector \(ATS\) workshop**](#)

2 Jul 10:30 | At CERN | Online | SCE | EN & FR
[**6th CERN Service Management Forum / 6è Forum "CERN Service Management"**](#)

2 Jul–2 Aug | Knowledge Sharing | Main Auditorium | Summer Student Programme | EN
[**Summer student lecture programme 2024**](#)

2 Jul–31 Jul | Knowledge Sharing | IT Amphitheatre | CERN openlab | EN
[**CERN openlab summer student lecture programme 2024**](#)

4 Jul | 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN
[**CERN70 public event: Exploring farther – Machines for new knowledge**](#)

6 Jul 15:00 | At CERN | R3 terrace, Prévessin | Music Club | EN & FR
[**Hardronic Music Festival 2024**](#)

Registration now open

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
[**Introduction to Accelerator Physics**](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN
[**DRD1 Gaseous Detectors School**](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

17 – 21 juin : Travaux sur la RD 884 (2X2 voies) entre le CERN et Thoiry

This announcement is only available in French at the moment. Thank you for your understanding.

Des travaux de réfection des enrobés sont planifiés **du 17 au 21 juin 2024** inclus sur la 2X2 voies. Ils se tiendront **sur les deux voies côté Alpes**, entre **l'échangeur de Badian et le rond-point de Porte de France**. Sur cette portion, la circulation se fera en double sens sur le côté Jura avec un basculement de la circulation à hauteur de la bretelle d'accès à la 2x2 voies depuis Val Thoiry direction Gex. Sur la portion à double sens, la vitesse sera limitée à 80 km/h. Le dispositif sera maintenu 24 h/24 durant les travaux.

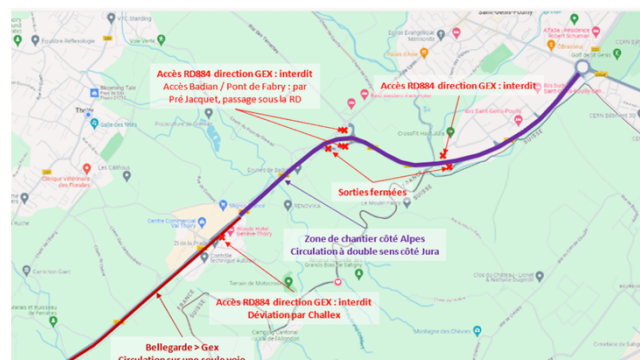
Attention !

- La circulation sera directement impactée **du 17 au 21 juin inclus**.
- **D'importants ralentissements sont à prévoir** sur la 2x2 voies à l'approche de Thoiry, et autour du giratoire de Val Thoiry, notamment en début de matinée.
- **Il ne sera pas possible de prendre la 2x2 voies à Val Thoiry en direction de Gex**. La déviation officielle est mise en place par la 2x2 voies, via l'échangeur de Challex.
- **Tous les accès et sorties dans le sens Bellegarde > Gex sont fermés entre Val Thoiry et Portes de France** (dernière entrée possible : Saint-Jean-de-Gonville, dernière sortie possible : Val Thoiry).
- La voie entre le rond-point des Terrettes et le petit rond-point de Badian est **fermée dans le sens Terrettes > Badian** : l'accès au

Pont de Fabry / écuries / voie cyclable se fait uniquement par le passage sous la 2x2 voies depuis Pré Jacquet.

- Tous les accès et sorties dans le sens Gex>Bellegarde sont maintenus.

Restez attentifs et prudents à l'approche de ces secteurs.



Plus d'informations sur le site web de la mairie de Thoiry: <https://www.mairie-thoiry.fr/travaux-sur-la-rd-884-2x2-voies/>

Parking outside Building 55: time restrictions apply

Since there are so few parking spaces around Building 55 and so many people coming to consult the services inside, parking in the car park next to

entrance B of the CERN site is **limited to a maximum of 30 minutes**. The car park is also **reserved for people visiting Buildings 55 and 33**.

As the signs indicate, a **parking disc** should be displayed in the front windscreen of vehicles parked there. Regular patrols are made by the Security service and, if the rules are not followed, offenders may be reported to the police and fined.

Thank you for your understanding.

SCE department

Calling all graduates and supervisors

You have until 25 June to give your feedback on the CERN Graduate programme

As part of its ongoing commitment to excellence, the Organization is launching a review of the CERN Graduate programme, just over one year after its start. But we can't do it without you. We need your insights, observations and suggestions. On 3 June, four surveys were sent by email to the three graduate populations (early career professionals, project graduates, research fellows), as well as to supervisors of graduates. Think of it as your opportunity to influence the future for graduates and their supervisors at CERN.

Why should you participate?

- **Make a difference:** Your feedback can lead to meaningful changes.
- **Voice your opinion:** This is your chance to share your story. What works well for you?

What could be improved? What should be changed?

- **Be heard:** No feedback is too small. Whether it's a suggestion for more social events, feedback on your work content or workload or a complete overhaul of how we recruit, onboard, supervise or train graduates, we want to hear it.

We know you're busy conquering the professional world, so we've made sure the survey won't take more than 15 minutes. Plus, your response will not be linked to you. **You have until 25 June to give your feedback** so grab a coffee, take a break and help us make our Graduate programme the best it can be.

HR department

Jardin des Particules: enrol your child for 2024/2025

It is still possible to enrol your child in the Jardin des Particules crèche or school for the 2024/2025 school year.

Le Jardin des Particules welcomes about 100 children, divided according to their age:

- The crèche welcomes children from four to 36 months (attendance two, three or five days per week);
- The school welcomes children aged between three and eight years (full day and attendance five days per week).

Enrol your child on the Jardin des Particules website: <https://nurseryschool.web.cern.ch/>.

You are welcome to attend Le Jardin des Particules **open days on 18 and 20 June, between 12.30 and 1.30 p.m.**

For more information, please contact:

- Carole Cherpin (enrolment/information): info.jdp@cern.ch
- Roberta Cavigliasso (management and educational information): roberta.cavigliasso@cern.ch

To book an individual appointment to visit the Jardin, please contact:

- info.jdp@cern.ch
- 0041 22 767 36 04 or 0041 75 411 34